Homework2 CNN实现SVHN图像分类任务

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一、任务说明与方法概览

SVHN (Street View House Numbers) 街景门牌数字数据集是一个真实世界图像数据集，来源于 Google 街景图像中的门牌号码。该数据集包含 Format1 和 Format2 两个组别： Format1 是原始数据图片，每张图片包含多个数字； Format2 则是在原始数据图片基础上，把每个数字单独裁剪成一张小图，只标注中心最主要的那个数字，且边上偶尔会有残影干扰，但每张图片仍然是单标签、单个数字。

本次实验选择 Format2 数据集，通过设计并训练卷积神经网络（CNN），实现对街景门牌数字的图像分类任务（0-9 共 10 类）。

二、实验目标

数据集结构：

图片规格为 32×32×3 RGB 彩色图像，包含数字 0~9 共10类。

数据规模：

训练集（train）：73257 张

测试集（test）：26032 张

开发环境：

PyTorch，torchvision，matplotlib

实现流程：

提示：CNN是作业基本要求，此外还可以引入 ResNet 等方式升级网络结构，也可以使用数据增强，从而提升分类准确率。

三、代码实现

In [1]:
```
import torch
import torch.nn as nn
import torch.optim as optim
from torch.utils.data import Dataset, DataLoader
from torchvision import transforms
import scipy.io as sio
```

```python
import numpy as np
import matplotlib.pyplot as plt

# ===================== 1. 全局超参数初始化 =====================
BATCH_SIZE = 64
EPOCHS = 30
LEARNING_RATE = 1e-3
NUM_CLASSES = 10
# 自动选择设备：优先GPU，无GPU则用CPU
DEVICE = torch.device("cuda" if torch.cuda.is_available() else "cpu")
# 数据集与模型保存路径（请将mat文件放在代码同目录下）
TRAIN_MAT_PATH = "c:/Users/boji/Desktop/homework1/homework2/train_32x32.mat"
TEST_MAT_PATH = "c:/Users/boji/Desktop/homework1/homework2/test_32x32.mat"
# 分别定义CNN和ResNet的保存路径
CNN_MODEL_SAVE_PATH = "./svhn_cnn_model.pth"
RESNET_MODEL_SAVE_PATH = "./svhn_resnet_model.pth"
CNN_CURVE_SAVE_PATH = "./svhn_cnn_training_curve.png"
RESNET_CURVE_SAVE_PATH = "./svhn_resnet_training_curve.png"

# ===================== 2. 自定义数据集类  =====================
# X(32,32,3,i)为第i张RGB图，y(i)为对应标签
class SVHNDataset(Dataset):
    def __init__(self, mat_file_path, transform=None):
        # 加载.mat文件
        mat_data = sio.loadmat(mat_file_path)

        # 维度转换：原始X shape为 (32,32,3,样本数) → 转为PyTorch要求的 (样本数,3,
        self.images = np.transpose(mat_data['X'], (3, 2, 0, 1))
        # 标签处理：SVHN原始标签中10代表数字0，转换为0-9的标准分类标签
        self.labels = mat_data['y'].flatten()
        self.labels[self.labels == 10] = 0

        self.transform = transform

    def __len__(self):
        return len(self.labels)

    def __getitem__(self, idx):
        # 获取单张图像和对应标签
        img = self.images[idx]  # 单张图shape: (3,32,32)
        label = self.labels[idx]

        # 转换为(H,W,C)格式适配transforms预处理
        img = img.transpose(1, 2, 0)
        if self.transform:
            img = self.transform(img)

        return img, torch.tensor(label, dtype=torch.long)

# ===================== 3. 数据预处理与数据增强 =====================
# 训练集：数据增强
train_transform = transforms.Compose([
    transforms.ToPILImage(),
    transforms.RandomCrop(32, padding=4),  # 随机裁剪
    transforms.RandomHorizontalFlip(p=0.5),  # 随机水平翻转
    transforms.ToTensor(),  # 转为tensor并归一化到[0,1]
    transforms.Normalize(mean=[0.485, 0.456, 0.406], std=[0.229, 0.224, 0.225])
])

# 测试集：仅基础预处理，不做数据增强
```

```
test_transform = transforms.Compose([
    transforms.ToPILImage(),
    transforms.ToTensor(),
    transforms.Normalize(mean=[0.485, 0.456, 0.406], std=[0.229, 0.224, 0.225])
])

# 构建数据集与数据加载器
train_dataset = SVHNDataset(TRAIN_MAT_PATH, transform=train_transform)
test_dataset = SVHNDataset(TEST_MAT_PATH, transform=test_transform)

train_loader = DataLoader(
    train_dataset, batch_size=BATCH_SIZE, shuffle=True, num_workers=0, pin_memor
)
test_loader = DataLoader(
    test_dataset, batch_size=BATCH_SIZE, shuffle=False, num_workers=0, pin_memor
)

# ==================== 4. 模型构建 ====================
# 基础CNN模型
class BaseCNN(nn.Module):
    def __init__(self, num_classes=10):
        super(BaseCNN, self).__init__()
        # 卷积特征提取层
        self.conv_block = nn.Sequential(
            # 第1卷积块
            nn.Conv2d(3, 16, kernel_size=3, stride=1, padding=1),
            nn.BatchNorm2d(16),
            nn.ReLU(inplace=True),
            nn.MaxPool2d(2, 2),  # 输出尺寸: 16×16×16

            # 第2卷积块
            nn.Conv2d(16, 32, kernel_size=3, stride=1, padding=1),
            nn.BatchNorm2d(32),
            nn.ReLU(inplace=True),
            nn.MaxPool2d(2, 2),  # 输出尺寸: 8×8×32

            # 第3卷积块
            nn.Conv2d(32, 64, kernel_size=3, stride=1, padding=1),
            nn.BatchNorm2d(64),
            nn.ReLU(inplace=True),
            nn.MaxPool2d(2, 2),  # 输出尺寸: 4×4×64
        )

        # 全连接分类层
        self.fc_block = nn.Sequential(
            nn.Dropout(0.5),  # 防止过拟合
            nn.Linear(64 * 4 * 4, 128),
            nn.ReLU(inplace=True),
            nn.Linear(128, num_classes)
        )

    def forward(self, x):
        x = self.conv_block(x)
        x = x.flatten(1)  # 展平特征
        x = self.fc_block(x)
        return x

# 拓展ResNet模型
from torchvision.models import resnet18
class ResNetSVHN(nn.Module):
```

```python
    def __init__(self, num_classes=10):
        super(ResNetSVHN, self).__init__()
        self.backbone = resnet18(pretrained=False)
        self.backbone.fc = nn.Linear(self.backbone.fc.in_features, num_classes)

    def forward(self, x):
        return self.backbone(x)

# ===================== 5. 训练与验证核心函数 =====================
# 单轮训练函数
def train_one_epoch(model, loader, criterion, optimizer, device):
    model.train()
    total_loss = 0.0
    total_correct = 0
    total_samples = 0

    for images, labels in loader:
        images, labels = images.to(device), labels.to(device)
        optimizer.zero_grad()  # 梯度清零

        # 前向传播
        outputs = model(images)
        loss = criterion(outputs, labels)

        # 反向传播与参数更新
        loss.backward()
        optimizer.step()

        # 统计指标
        total_loss += loss.item() * images.shape[0]
        _, preds = torch.max(outputs, 1)
        total_correct += (preds == labels).sum().item()
        total_samples += images.shape[0]

    epoch_loss = total_loss / total_samples
    epoch_acc = total_correct / total_samples
    return epoch_loss, epoch_acc

# 单轮测试/验证函数
def test_one_epoch(model, loader, criterion, device):
    model.eval()
    total_loss = 0.0
    total_correct = 0
    total_samples = 0

    with torch.no_grad():  # 关闭梯度计算
        for images, labels in loader:
            images, labels = images.to(device), labels.to(device)
            outputs = model(images)
            loss = criterion(outputs, labels)

            # 统计指标
            total_loss += loss.item() * images.shape[0]
            _, preds = torch.max(outputs, 1)
            total_correct += (preds == labels).sum().item()
            total_samples += images.shape[0]

    epoch_loss = total_loss / total_samples
    epoch_acc = total_correct / total_samples
    return epoch_loss, epoch_acc
```

```python
# ===================== 6. 通用训练函数 =====================
def train_model(model, model_name, model_save_path, curve_save_path):
    """
    通用训练函数
    :param model: 要训练的模型
    :param model_name: 模型名称（用于打印日志）
    :param model_save_path: 模型保存路径
    :param curve_save_path: 训练曲线保存路径
    :return: 训练历史记录
    """
    # 初始化损失函数、优化器、学习率调度器
    criterion = nn.CrossEntropyLoss()  # 多分类标准损失函数
    optimizer = optim.Adam(model.parameters(), lr=LEARNING_RATE)
    scheduler = optim.lr_scheduler.StepLR(optimizer, step_size=5, gamma=0.5)  #

    # 记录训练过程指标
    train_loss_history = []
    train_acc_history = []
    test_loss_history = []
    test_acc_history = []

    print(f"\n{'='*60}")
    print(f"开始训练 {model_name} 模型，运行设备：{DEVICE}")
    print(f"{'='*60}")

    # 迭代训练
    for epoch in range(EPOCHS):
        train_loss, train_acc = train_one_epoch(model, train_loader, criterion,
        test_loss, test_acc = test_one_epoch(model, test_loader, criterion, DEVI
        scheduler.step()  # 更新学习率

        # 保存历史指标
        train_loss_history.append(train_loss)
        train_acc_history.append(train_acc)
        test_loss_history.append(test_loss)
        test_acc_history.append(test_acc)

        # 打印训练日志
        print(f"{model_name} - Epoch [{epoch+1}/{EPOCHS}]")
        print(f"训练损失：{train_loss:.4f} | 训练准确率：{train_acc:.4f}")
        print(f"测试损失：{test_loss:.4f} | 测试准确率：{test_acc:.4f}")
        print("-"*60)

    # 保存训练好的模型
    torch.save(model.state_dict(), model_save_path)
    print(f"\n{model_name} 训练完成！模型已保存至：{model_save_path}")
    print(f"{model_name} 最终测试集准确率：{test_acc_history[-1]:.4f}")

    # 绘制并保存训练曲线
    plot_training_curve(train_loss_history, train_acc_history,
                        test_loss_history, test_acc_history,
                        model_name, curve_save_path)

    # 返回训练历史，方便后续对比
    return {
        'train_loss': train_loss_history,
        'train_acc': train_acc_history,
        'test_loss': test_loss_history,
        'test_acc': test_acc_history
```

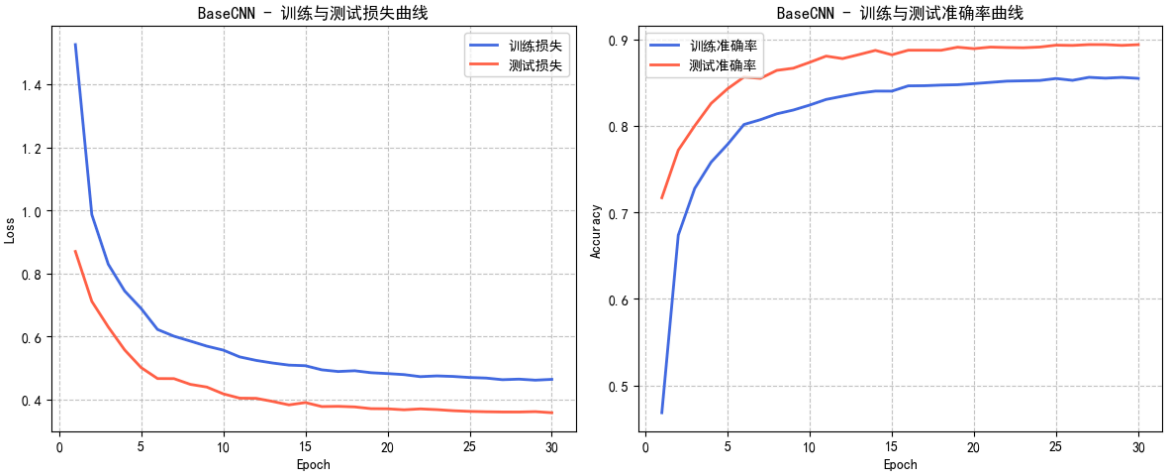
```
    }

# ===================== 7. 绘图函数 =====================
def plot_training_curve(train_loss, train_acc, test_loss, test_acc, model_name,
    """绘制训练曲线"""
    plt.rcParams['font.sans-serif'] = ['SimHei']
    plt.rcParams['axes.unicode_minus'] = False

    # 绘制损失曲线与准确率曲线
    fig, (ax1, ax2) = plt.subplots(1, 2, figsize=(12, 5))

    # 损失曲线
    ax1.plot(range(1, EPOCHS+1), train_loss, label='训练损失', color='royalblue'
    ax1.plot(range(1, EPOCHS+1), test_loss, label='测试损失', color='tomato', li
    ax1.set_xlabel('Epoch')
    ax1.set_ylabel('Loss')
    ax1.set_title(f'{model_name} - 训练与测试损失曲线')
    ax1.legend()
    ax1.grid(True, linestyle='--', alpha=0.7)

    # 准确率曲线
    ax2.plot(range(1, EPOCHS+1), train_acc, label='训练准确率', color='royalblue
    ax2.plot(range(1, EPOCHS+1), test_acc, label='测试准确率', color='tomato', l
    ax2.set_xlabel('Epoch')
    ax2.set_ylabel('Accuracy')
    ax2.set_title(f'{model_name} - 训练与测试准确率曲线')
    ax2.legend()
    ax2.grid(True, linestyle='--', alpha=0.7)

    plt.tight_layout()
    plt.savefig(save_path, dpi=300, bbox_inches='tight')
    plt.show()

# ===================== 8. 主函数：分别训练CNN和ResNet =====================
if __name__ == "__main__":
    # 1. 训练基础CNN模型
    cnn_model = BaseCNN(NUM_CLASSES).to(DEVICE)
    cnn_history = train_model(
        model=cnn_model,
        model_name="BaseCNN",
        model_save_path=CNN_MODEL_SAVE_PATH,
        curve_save_path=CNN_CURVE_SAVE_PATH
    )

    # 2. 训练ResNet模型
    resnet_model = ResNetSVHN(NUM_CLASSES).to(DEVICE)
    resnet_history = train_model(
        model=resnet_model,
        model_name="ResNet18",
        model_save_path=RESNET_MODEL_SAVE_PATH,
        curve_save_path=RESNET_CURVE_SAVE_PATH
    )

    # 3. 打印最终对比结果
    print(f"\n{'='*60}")
    print("模型训练结果对比")
    print(f"{'='*60}")
    print(f"BaseCNN 最终测试准确率: {cnn_history['test_acc'][-1]:.4f}")
    print(f"ResNet18 最终测试准确率: {resnet_history['test_acc'][-1]:.4f}")
    print(f"{'='*60}")
```

```
# 绘制对比曲线
plt.rcParams['font.sans-serif'] = ['SimHei']
plt.rcParams['axes.unicode_minus'] = False
fig, (ax1, ax2) = plt.subplots(1, 2, figsize=(12, 5))

# 损失对比
ax1.plot(range(1, EPOCHS+1), cnn_history['test_loss'], label='CNN测试损失',
ax1.plot(range(1, EPOCHS+1), resnet_history['test_loss'], label='ResNet测试
ax1.set_xlabel('Epoch')
ax1.set_ylabel('Loss')
ax1.set_title('CNN vs ResNet 测试损失对比')
ax1.legend()
ax1.grid(True, linestyle='--', alpha=0.7)

# 准确率对比
ax2.plot(range(1, EPOCHS+1), cnn_history['test_acc'], label='CNN测试准确率',
ax2.plot(range(1, EPOCHS+1), resnet_history['test_acc'], label='ResNet测试准
ax2.set_xlabel('Epoch')
ax2.set_ylabel('Accuracy')
ax2.set_title('CNN vs ResNet 测试准确率对比')
ax2.legend()
ax2.grid(True, linestyle='--', alpha=0.7)

plt.tight_layout()
plt.savefig("./svhn_cnn_vs_resnet.png", dpi=300, bbox_inches='tight')
plt.show()
```

```
============================================================
开始训练 BaseCNN 模型，运行设备：cuda
============================================================
BaseCNN - Epoch [1/30]
训练损失: 1.5264 | 训练准确率: 0.4687
测试损失: 0.8693 | 测试准确率: 0.7168
------------------------------------------------------------
BaseCNN - Epoch [2/30]
训练损失: 0.9864 | 训练准确率: 0.6736
测试损失: 0.7108 | 测试准确率: 0.7718
------------------------------------------------------------
BaseCNN - Epoch [3/30]
训练损失: 0.8289 | 训练准确率: 0.7276
测试损失: 0.6296 | 测试准确率: 0.7999
------------------------------------------------------------
BaseCNN - Epoch [4/30]
训练损失: 0.7437 | 训练准确率: 0.7580
测试损失: 0.5564 | 测试准确率: 0.8258
------------------------------------------------------------
BaseCNN - Epoch [5/30]
训练损失: 0.6882 | 训练准确率: 0.7785
测试损失: 0.5005 | 测试准确率: 0.8427
------------------------------------------------------------
BaseCNN - Epoch [6/30]
训练损失: 0.6216 | 训练准确率: 0.8014
测试损失: 0.4656 | 测试准确率: 0.8563
------------------------------------------------------------
BaseCNN - Epoch [7/30]
训练损失: 0.6002 | 训练准确率: 0.8070
测试损失: 0.4654 | 测试准确率: 0.8547
------------------------------------------------------------
BaseCNN - Epoch [8/30]
训练损失: 0.5846 | 训练准确率: 0.8138
测试损失: 0.4472 | 测试准确率: 0.8639
------------------------------------------------------------
BaseCNN - Epoch [9/30]
训练损失: 0.5686 | 训练准确率: 0.8181
测试损失: 0.4386 | 测试准确率: 0.8664
------------------------------------------------------------
BaseCNN - Epoch [10/30]
训练损失: 0.5560 | 训练准确率: 0.8239
测试损失: 0.4171 | 测试准确率: 0.8732
------------------------------------------------------------
BaseCNN - Epoch [11/30]
训练损失: 0.5344 | 训练准确率: 0.8304
测试损失: 0.4034 | 测试准确率: 0.8804
------------------------------------------------------------
BaseCNN - Epoch [12/30]
训练损失: 0.5233 | 训练准确率: 0.8341
测试损失: 0.4029 | 测试准确率: 0.8775
------------------------------------------------------------
BaseCNN - Epoch [13/30]
训练损失: 0.5150 | 训练准确率: 0.8376
测试损失: 0.3930 | 测试准确率: 0.8822
------------------------------------------------------------
BaseCNN - Epoch [14/30]
训练损失: 0.5083 | 训练准确率: 0.8400
测试损失: 0.3820 | 测试准确率: 0.8870
------------------------------------------------------------
BaseCNN - Epoch [15/30]
```

```
训练损失: 0.5064 | 训练准确率: 0.8400
测试损失: 0.3894 | 测试准确率: 0.8819
------------------------------------------------------------
BaseCNN - Epoch [16/30]
训练损失: 0.4933 | 训练准确率: 0.8461
测试损失: 0.3767 | 测试准确率: 0.8872
------------------------------------------------------------
BaseCNN - Epoch [17/30]
训练损失: 0.4880 | 训练准确率: 0.8462
测试损失: 0.3774 | 测试准确率: 0.8872
------------------------------------------------------------
BaseCNN - Epoch [18/30]
训练损失: 0.4904 | 训练准确率: 0.8469
测试损失: 0.3756 | 测试准确率: 0.8871
------------------------------------------------------------
BaseCNN - Epoch [19/30]
训练损失: 0.4841 | 训练准确率: 0.8474
测试损失: 0.3700 | 测试准确率: 0.8907
------------------------------------------------------------
BaseCNN - Epoch [20/30]
训练损失: 0.4814 | 训练准确率: 0.8487
测试损失: 0.3696 | 测试准确率: 0.8890
------------------------------------------------------------
BaseCNN - Epoch [21/30]
训练损失: 0.4782 | 训练准确率: 0.8501
测试损失: 0.3664 | 测试准确率: 0.8908
------------------------------------------------------------
BaseCNN - Epoch [22/30]
训练损失: 0.4716 | 训练准确率: 0.8515
测试损失: 0.3693 | 测试准确率: 0.8903
------------------------------------------------------------
BaseCNN - Epoch [23/30]
训练损失: 0.4740 | 训练准确率: 0.8519
测试损失: 0.3670 | 测试准确率: 0.8901
------------------------------------------------------------
BaseCNN - Epoch [24/30]
训练损失: 0.4723 | 训练准确率: 0.8523
测试损失: 0.3635 | 测试准确率: 0.8909
------------------------------------------------------------
BaseCNN - Epoch [25/30]
训练损失: 0.4690 | 训练准确率: 0.8546
测试损失: 0.3614 | 测试准确率: 0.8931
------------------------------------------------------------
BaseCNN - Epoch [26/30]
训练损失: 0.4672 | 训练准确率: 0.8524
测试损失: 0.3602 | 测试准确率: 0.8927
------------------------------------------------------------
BaseCNN - Epoch [27/30]
训练损失: 0.4620 | 训练准确率: 0.8559
测试损失: 0.3595 | 测试准确率: 0.8937
------------------------------------------------------------
BaseCNN - Epoch [28/30]
训练损失: 0.4638 | 训练准确率: 0.8549
测试损失: 0.3594 | 测试准确率: 0.8937
------------------------------------------------------------
BaseCNN - Epoch [29/30]
训练损失: 0.4605 | 训练准确率: 0.8558
测试损失: 0.3606 | 测试准确率: 0.8928
------------------------------------------------------------
BaseCNN - Epoch [30/30]
```

```
训练损失: 0.4631 | 训练准确率: 0.8547
测试损失: 0.3571 | 测试准确率: 0.8936
------------------------------------------------------------

BaseCNN 训练完成！模型已保存至: ./svhn_cnn_model.pth
BaseCNN 最终测试集准确率: 0.8936
```

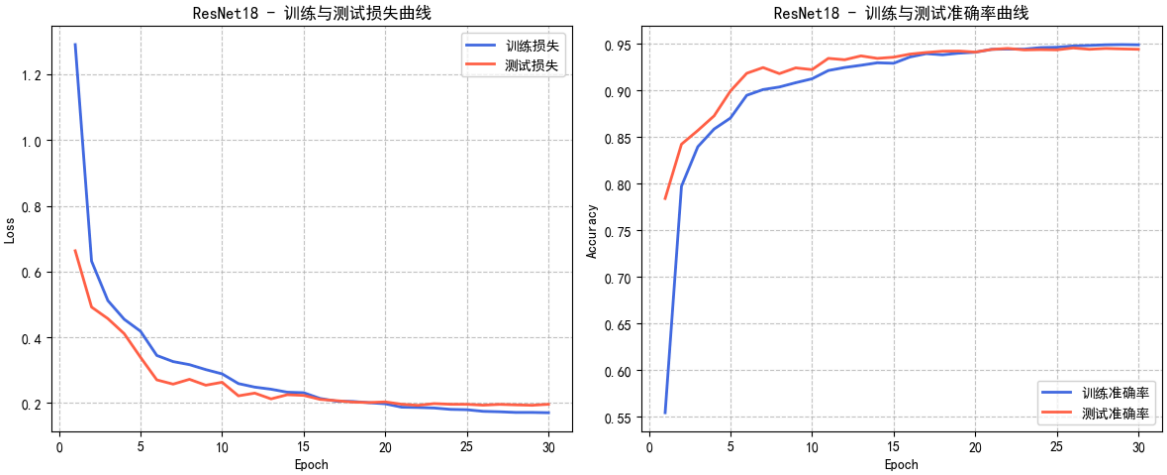



```
c:\Users\boji\AppData\Local\Programs\Python\Python38\lib\site-packages\torchvisio
n\models\_utils.py:208: UserWarning: The parameter 'pretrained' is deprecated sin
ce 0.13 and may be removed in the future, please use 'weights' instead.
  warnings.warn(
c:\Users\boji\AppData\Local\Programs\Python\Python38\lib\site-packages\torchvisio
n\models\_utils.py:223: UserWarning: Arguments other than a weight enum or `None`
for 'weights' are deprecated since 0.13 and may be removed in the future. The cur
rent behavior is equivalent to passing `weights=None`.
  warnings.warn(msg)
```

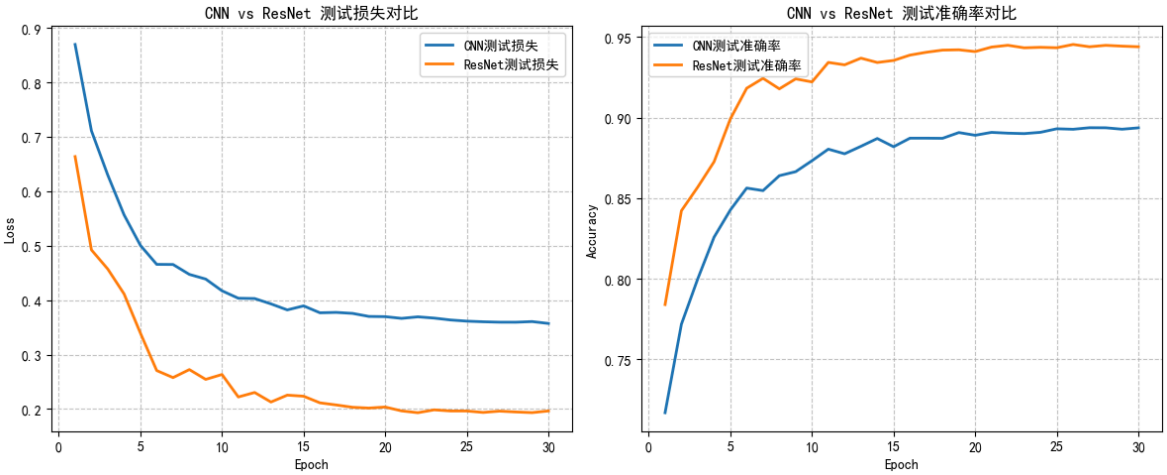
```
============================================================
开始训练 ResNet18 模型，运行设备：cuda
============================================================
ResNet18 - Epoch [1/30]
训练损失: 1.2908 | 训练准确率: 0.5543
测试损失: 0.6635 | 测试准确率: 0.7840
------------------------------------------------------------
ResNet18 - Epoch [2/30]
训练损失: 0.6320 | 训练准确率: 0.7971
测试损失: 0.4922 | 测试准确率: 0.8422
------------------------------------------------------------
ResNet18 - Epoch [3/30]
训练损失: 0.5119 | 训练准确率: 0.8394
测试损失: 0.4571 | 测试准确率: 0.8569
------------------------------------------------------------
ResNet18 - Epoch [4/30]
训练损失: 0.4551 | 训练准确率: 0.8585
测试损失: 0.4113 | 测试准确率: 0.8726
------------------------------------------------------------
ResNet18 - Epoch [5/30]
训练损失: 0.4187 | 训练准确率: 0.8702
测试损失: 0.3395 | 测试准确率: 0.8992
------------------------------------------------------------
ResNet18 - Epoch [6/30]
训练损失: 0.3450 | 训练准确率: 0.8946
测试损失: 0.2707 | 测试准确率: 0.9182
------------------------------------------------------------
ResNet18 - Epoch [7/30]
训练损失: 0.3263 | 训练准确率: 0.9008
测试损失: 0.2577 | 测试准确率: 0.9244
------------------------------------------------------------
ResNet18 - Epoch [8/30]
训练损失: 0.3170 | 训练准确率: 0.9035
测试损失: 0.2724 | 测试准确率: 0.9178
------------------------------------------------------------
ResNet18 - Epoch [9/30]
训练损失: 0.3019 | 训练准确率: 0.9081
测试损失: 0.2545 | 测试准确率: 0.9240
------------------------------------------------------------
ResNet18 - Epoch [10/30]
训练损失: 0.2889 | 训练准确率: 0.9123
测试损失: 0.2634 | 测试准确率: 0.9221
------------------------------------------------------------
ResNet18 - Epoch [11/30]
训练损失: 0.2594 | 训练准确率: 0.9212
测试损失: 0.2221 | 测试准确率: 0.9342
------------------------------------------------------------
ResNet18 - Epoch [12/30]
训练损失: 0.2486 | 训练准确率: 0.9244
测试损失: 0.2304 | 测试准确率: 0.9327
------------------------------------------------------------
ResNet18 - Epoch [13/30]
训练损失: 0.2423 | 训练准确率: 0.9268
测试损失: 0.2129 | 测试准确率: 0.9369
------------------------------------------------------------
ResNet18 - Epoch [14/30]
训练损失: 0.2332 | 训练准确率: 0.9295
测试损失: 0.2255 | 测试准确率: 0.9342
------------------------------------------------------------
ResNet18 - Epoch [15/30]
```

```
训练损失: 0.2316 | 训练准确率: 0.9290
测试损失: 0.2235 | 测试准确率: 0.9355
------------------------------------------------------------
ResNet18 - Epoch [16/30]
训练损失: 0.2142 | 训练准确率: 0.9357
测试损失: 0.2114 | 测试准确率: 0.9388
------------------------------------------------------------
ResNet18 - Epoch [17/30]
训练损失: 0.2059 | 训练准确率: 0.9393
测试损失: 0.2075 | 测试准确率: 0.9405
------------------------------------------------------------
ResNet18 - Epoch [18/30]
训练损失: 0.2050 | 训练准确率: 0.9381
测试损失: 0.2032 | 测试准确率: 0.9419
------------------------------------------------------------
ResNet18 - Epoch [19/30]
训练损失: 0.2013 | 训练准确率: 0.9398
测试损失: 0.2018 | 测试准确率: 0.9421
------------------------------------------------------------
ResNet18 - Epoch [20/30]
训练损失: 0.1980 | 训练准确率: 0.9408
测试损失: 0.2037 | 测试准确率: 0.9410
------------------------------------------------------------
ResNet18 - Epoch [21/30]
训练损失: 0.1877 | 训练准确率: 0.9439
测试损失: 0.1966 | 测试准确率: 0.9438
------------------------------------------------------------
ResNet18 - Epoch [22/30]
训练损失: 0.1867 | 训练准确率: 0.9442
测试损失: 0.1932 | 测试准确率: 0.9449
------------------------------------------------------------
ResNet18 - Epoch [23/30]
训练损失: 0.1852 | 训练准确率: 0.9441
测试损失: 0.1985 | 测试准确率: 0.9433
------------------------------------------------------------
ResNet18 - Epoch [24/30]
训练损失: 0.1810 | 训练准确率: 0.9457
测试损失: 0.1964 | 测试准确率: 0.9436
------------------------------------------------------------
ResNet18 - Epoch [25/30]
训练损失: 0.1799 | 训练准确率: 0.9461
测试损失: 0.1964 | 测试准确率: 0.9433
------------------------------------------------------------
ResNet18 - Epoch [26/30]
训练损失: 0.1750 | 训练准确率: 0.9475
测试损失: 0.1937 | 测试准确率: 0.9454
------------------------------------------------------------
ResNet18 - Epoch [27/30]
训练损失: 0.1737 | 训练准确率: 0.9481
测试损失: 0.1962 | 测试准确率: 0.9440
------------------------------------------------------------
ResNet18 - Epoch [28/30]
训练损失: 0.1716 | 训练准确率: 0.9488
测试损失: 0.1946 | 测试准确率: 0.9448
------------------------------------------------------------
ResNet18 - Epoch [29/30]
训练损失: 0.1716 | 训练准确率: 0.9490
测试损失: 0.1934 | 测试准确率: 0.9443
------------------------------------------------------------
ResNet18 - Epoch [30/30]
```

```
训练损失: 0.1708 | 训练准确率: 0.9488
测试损失: 0.1965 | 测试准确率: 0.9440
------------------------------------------------------------

ResNet18 训练完成! 模型已保存至: ./svhn_resnet_model.pth
ResNet18 最终测试集准确率: 0.9440
```




```
============================================================
模型训练结果对比
============================================================
BaseCNN 最终测试准确率: 0.8936
ResNet18 最终测试准确率: 0.9440
============================================================
```



四、实验结果与对比

BaseCNN最终正确率为89.36%，收敛稳定，满足基础分类需求；ResNet18最终正确率为94.40%，收敛更快、精度更高，泛化能力更强。

实验结果表明，ResNet18 相比基础 CNN，测试准确率提升约5 个百分点，在复杂背景与干扰信息下的数字识别效果更优。训练曲线显示，两种模型的损失持续下降、准确率稳步上升，无明显过拟合，训练过程合理有效。

五、实验结论

基础 CNN 可有效完成 SVHN 数字分类任务，达到课程作业要求。残差结构能显著提升网络性能，ResNet18 在准确率与收敛速度上均优于基础 CNN。数据增强、批归一化、学

习率衰减等策略，有效提升模型泛化能力与训练稳定性。本次实验完整复现了图像分类全流程，代码可运行、结果可复现，成功达成实验目标。